Unit III

What is Bulk Deformation?

Metal forming operations which cause significant shape change by plastic deformation in metallic parts are referred to *bulk deformation processes*. In most of the cases cylindrical bars and billets, rectangular billets and slabs, and similar shapes are the initial shapes which are plastically deformed in cold, warm or hot conditions into a desired shape.

Complex shapes with good mechanical properties can be produced by bulk deformation processes. Following metal forming operations are commonly referred to as *bulk deformation processes*.

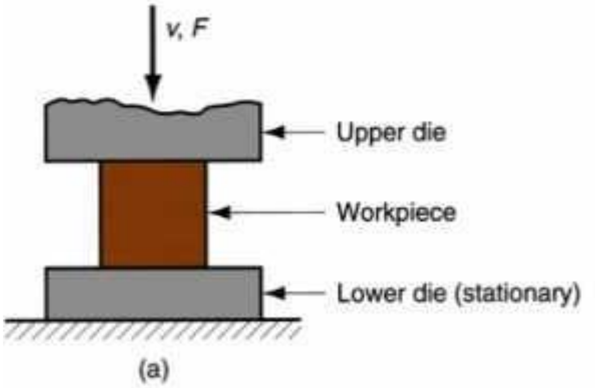
1. ***Forging***– The *initial shape* is squeezed and shaped between two opposing dies.
2. ***Rolling***– A *slab* or *plate* is squeezed between two rotating rolls to reduce the height so as to produce thinner shapes like *sheets*,
3. ***Extrusion***– The *initial shape* is squeezed through a shaped die such that the cross-section of the deformed part becomes similar to that of the die opening.
4. ***Wire and bar drawing***– The diameter of typical *cylindrical bars* or *wires* is reduced by pulling it through a shaped die.

FORGING

Forging is the process by which a metallic part is deformed to a final shape with the application of pressure and with or without the application of heat. Forging processes can broadly be classified as follows.

Open Die Forging

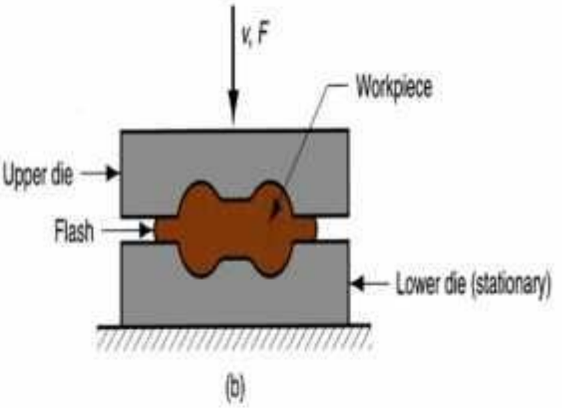
The work piece is compressed between two flat dies facilitating lateral flow of material without constraint,



(a)

Impression Die Forging

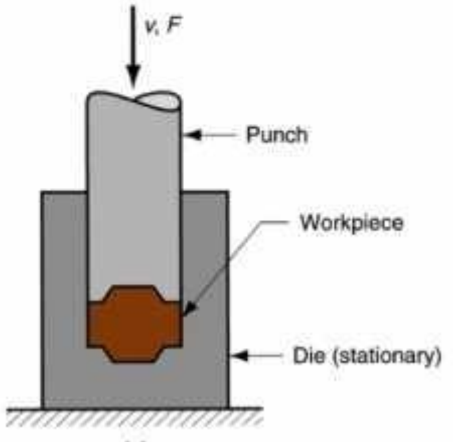
The work piece is compressed between two dies with a cavity or impression that is imparted to the work piece. The metal flow is constrained within the impression of the dies. However, the excess material remains as *flash*.



(b)

Flashless Forging

This is an improved version of *Impression-die forging* process. The initial volume of the workpiece is carefully taken so that no excess *flash* is produced.



(c)

ADVANTAGES OF *FORGING PROCESSES*.

(1) It improves the structure as well as mechanical properties of the metallic parts.

(2) Forging facilitates orientation of grains in a desired direction to improve the mechanical properties.

(3) Forged parts are consistent in shape with the minimum presence of voids and porosities.

(4) Forging can produce parts with high strength to weight ratio.

(5) Forging processes are very economical for moderate to high volume productions.

DEFECTS IN FORGING

Though forging process provides good quality products compared to other manufacturing processes, some defects that are lightly to come if proper care is not taken in forging process design. A brief description of such defects and their remedial method is given below.

- **Unfilled Section**

This defect refers to localized unfilled portion within the die cavity due to improper design of the forging die or inappropriate selection of the forging technique.

- **Cold Shut**

Cold shut appears as small cracks at the corners of the forged part that is caused primarily due to very tight fillet radii that inhibit smooth material flow towards the corner of the die.

- **Scale Pits**

Scale pits are seen as irregular deputations on the surface of the forging that is caused primarily due to improper cleaning of the stock used for forging. The oxide and scale gets embedded into the finish forging surface. When the forging is cleaned by pickling, these are seen as deputations on the forging surface.

- **Die Shift**

Die shift is caused by the misalignment of the top and the bottom dies making the two halves of the forging to be improper shape.

- **Flakes**

These are basically internal ruptures caused by the improper cooling of the large forging. Rapid cooling causes the exterior to cool quickly causing internal fractures. This can be remedied by following proper cooling practices.

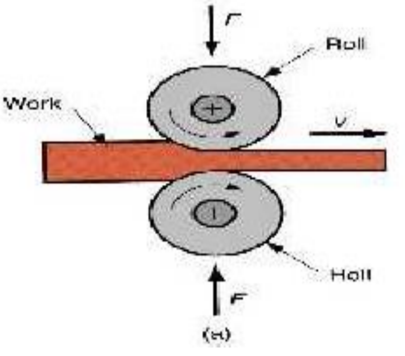
- **Improper Grain Flow**

This is primarily caused by the improper design of the die that induces material flow in an inappropriate manner leading to various defects.

ROLLING

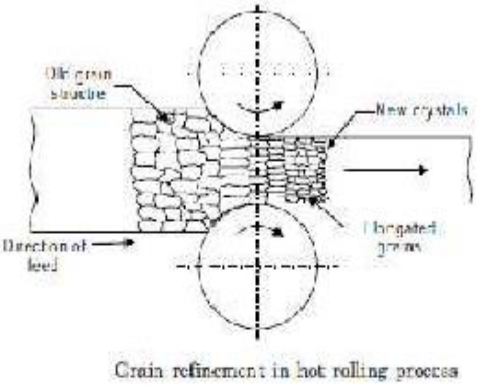
Rolling is a forming operation where cylindrical rolls are used to reduce the cross sectional area of a bar or plate with a corresponding increase in the length. *Rolling process* is widely used because of high productivity.

Rolling processes are broadly classified by the geometry of the final rolled shape of the workpiece material such as *flat rolling* that is used to reduce thickness of a rectangular.



Hot Rolling*-Rolling processes* are performed both at high temperature above the *recrystallization temperature*.

Rolling is the most rapid method of forming metal into desired shapes by plastic deformationthrough compressive stresses using two or more than two rolls. It is one of the most widelyused of all the metal working processes.



Grain refinement in hot rolling process

The main objective of rolling is to convert largersections such as ingots into smaller sections which can be used either directly in as rolledstate or as stock for working through other processes. The coarse structure of cast ingot isconvened into a fine grained structure using rolling process

Cold Rolling-*Rolling processes* are performed both at room temperature.

Hot Rolling is usually performed when large amount of deformation is required while *Cold Rolling* is performed for finished sheet and plate stock. Various structural members, plates and sheets as well as pipes are produced by rolling at very high productivity although due to high tooling cost, it is economical for large batch size only.

CLASSIFICATION OF ROLLING

Two-High Rolling Mill

A two-high rolling mill has two horizontal rolls revolving at the same speed butin opposite direction. The rolls are supported on bearings housed in sturdy upright sideframes called stands. The space between the rolls can be adjusted by raising or 1owering theupper roll. Their direction of rotation is fixed and cannot be reversed. The reduction in thethickness of work is achieved by feeding from one direction only.

Two-high rolling mill reverse

which incorporates a drive mechanism that can reverse thedirection of rotation of the rolls. In a two-high reversing rolling mill, there is continuous rolling of the work piece throughback-and-forth passes between the rolls.

Three-High Rolling Mills

It consists of three parallel rolls, arranged one above the other as shown .Thedirections of rotation of the upper and lower rolls are the same but the intermediate rollrotates in a direction opposite to both of these. This type of rolling mill is used for rollingof two continuous passes in a rolling sequence without reversing the drives. This results ina higher rate of production than the two-high rolling mill.

Four-High Rolling Mill

It is essentially a two-high rolling mill, but with small sized rolls. Practically, it consists offour horizontal rolls, the two middle rolls are smaller in size than the top and bottom rolls
As shown the smaller size rolls are known as working rolls which concentratethe total rolling pressure over the work piece.

The larger diameter rolls are called back-uprolls and their main function is to prevent the deflection of the smaller rolls, which otherwisewould result in thickening of rolled plates or sheets at the centre. The common products ofthese mills are hot or cold rolled plates and sheets.

Cluster Mill

It is a special type of four-high rolling mill in which each of the two smaller working rollsare backed up by two or more of the larger back-up rolls as shown .. This type of mill is generally used for cold rolling work.

Continuous Rolling Mill

It consists of a number of non-reversing two-high rolling mills arranged one after the other,so that the material can be passed through all of them in sequence. It is suitable for massProduction work only, because for smaller quantities quick changes of set-up will be required
and they will consume lot of time and labor.

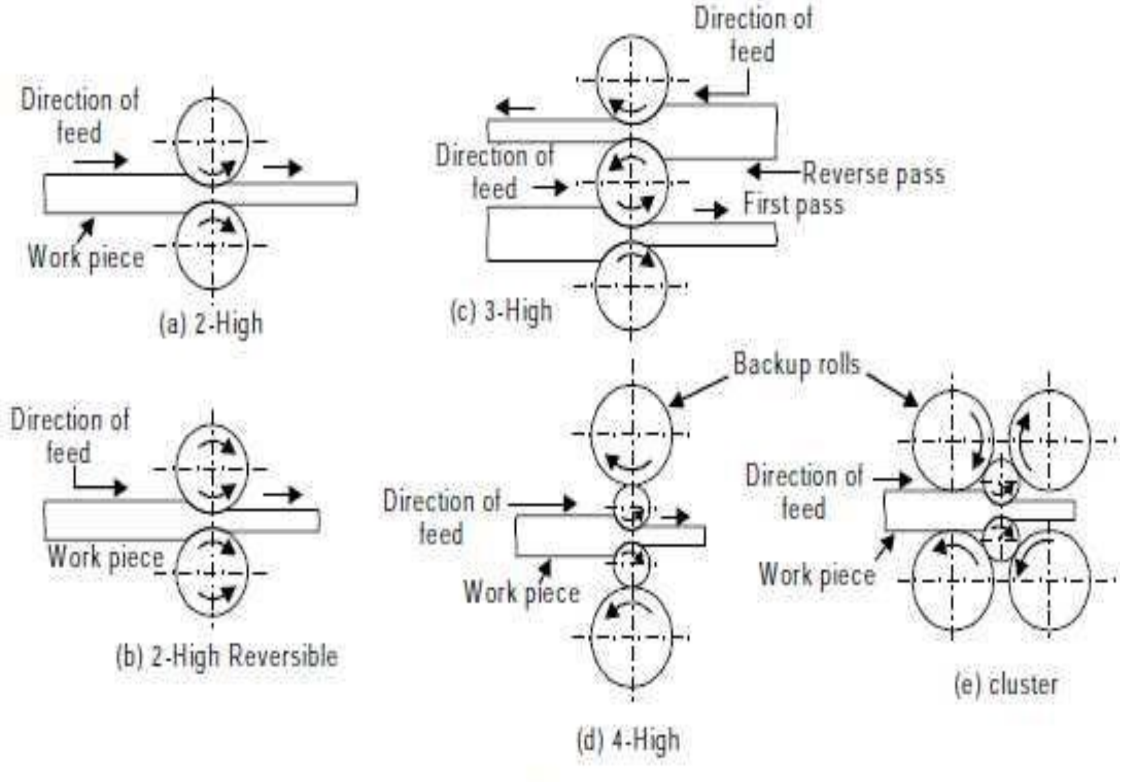


Fig. 15.2 Hot rolling stand arrangements

DEFECTS IN ROLLING

The defects in rolling can be classified as (a) surface defects, and (b) structural defects.

- **The surface defects**- include rusting and scaling, scratches and cracks on the surface, pits left on the surface due to subsequent detachment or removal of scales.
- **The structural defects-** are more important rolling defects some of which are difficult to remove. Some common structural defects in rolling are as follows.

- **Wavy edges and zipper cracks**

    These defects are caused due to bending of rolls under the rolling pressure

Edge cracks and centre split These defects are caused due to non-homogeneous plastic deformation of metal across the width **Alligatoring**

- **Folds**

    This defect is encountered when the reduction per pass is very low.

- **Laminations**

    Laminations mean layers. If the ingot is not sound and has a piping or blow holes and during rolling they do not get completely welded it will cause a defect called laminations.

Rolling defects (a) zipper crack, (b) Edge crack (c) Alligatoring

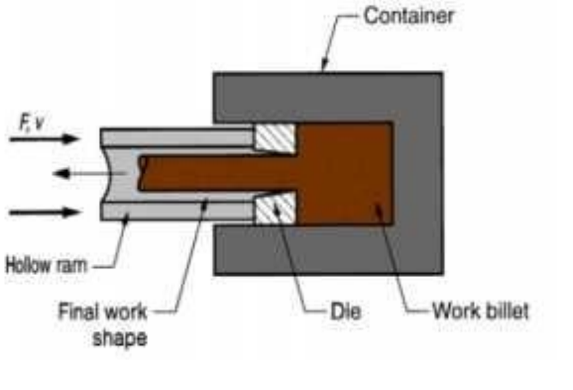
EXTRUSION

Extrusion is the process in which the work piece material is forced to flow through a die opening by applying compressive force to produce a desired cross-sectional shape. In general, extrusion is used to produce long parts of uniform cross-sections.

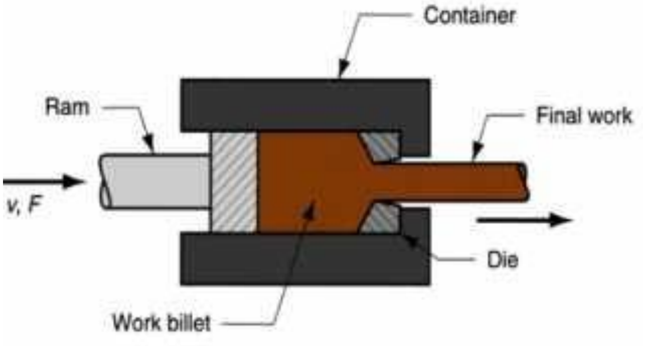
Direct and Indirect extrusions are two commonly used extrusion processes.

Extrusion processes are also performed at temperature well above the recrystallization temperature of the work piece material and at room temperature, which are referred to as hot and cold extrusion.

In direct extrusion-the extruded product moves in the same direction of the movement of the punch. As a result the billet has to overcome the frictional resistance of the container wall increasing the power requirement.



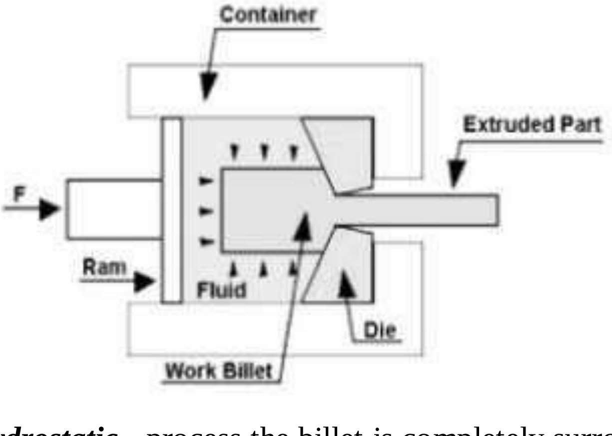
In indirect extrusion- the billet remains stationary inside the container and the extruded part comes out in the direction opposite to the punch travel. The power requirement in indirect extrusion is lesser compared to the same in direct extrusion as the frictional force is absent in indirect absence due to stationary billet.

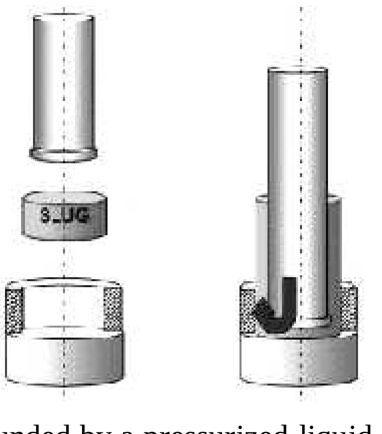


The principle of extrusion using a mandrel is similar to that of direct extrusion. Here the mandrel provides proper guidance and support for material flow.

In *impact extrusion*- the punch travels at a high speed and strikes the blank extruding it upwards. The metal blank deforms to fit the punch on the inside and the die on the outside.

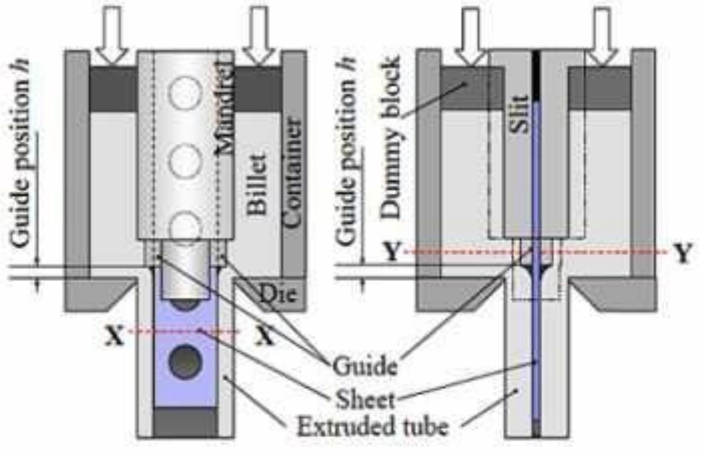
Lubricants are also added to aid the machines benefit for an easier punch-out. It only takes one impact for the finished shape to form from the blank. Once the blank has been contoured to the desired shape, a counter-punch ejector removes the work piece from within the die.





In the *hydrostatic* - process the billet is completely surrounded by a pressurized liquid, except where the billet contacts the die. This process can be done hot, warm, or cold, however the temperature is limited by the stability of the fluid used.

Although the presence of fluid inside the extrusion chamber eliminates the container wall friction, the process finds limited application because of need for special equipment and tooling.



ADVANTAGES OF EXTRUSION PROCESSES.

(1) *Extrusion* can produce variety of shapes with uniform cross-section.

(2) The grain structure and mechanical strength of workpiece material are improved in cold and warm extrusion processes.

(3) Cold extrusion can provide close tolerances.

(4) Wastage of material is the minimum in extrusion processes.

(5) Extrusion can be performed even for relatively brittle materials.

DEFECTS IN EXTRUSION

- **Surface cracking**

Surface cracking occurs when the surface of an extrusion splits, which is often caused by the extrusion temperature, friction, or speed being too high. It can also happen at lower temperatures if the extruded product temporarily sticks to the die.

- **Internal cracking**

Internal cracking occurs when the centre of the extrusion develops cracks or voids. These cracks are attributed to a state of hydrostatic tensile stress at the center line in the deformation zone in the die.

- **Pipe**

It is the flow pattern that draws the surface oxides and impurities to the centre of the product. Such a pattern is often caused by high friction or cooling of the outer regions of the billet.

- **Surface lines**

These are the lines visible on the surface of the extruded profile. This depends heavily on the quality of the die production and how well the die is maintained, as some residues of the material extruded can stick to the die surface and produce the embossed lines.

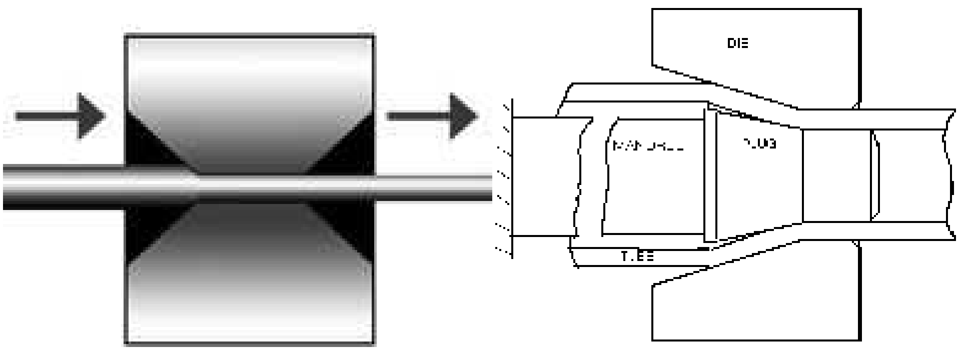
WIRE AND TUBE DRAWING

Wire drawingis a metalworking process used to reduce the cross-section of a wire by pulling the wire through a single, or series of, drawing die(s).

Although similar in process, **drawing** is different from *extrusion*, because in drawing the wire is pulled, rather than pushed, through the die.

Drawing is usually performed at **room temperature**, thus classified as a cold working process, but it may be performed at elevated temperatures for large wires to reduce forces.

The concept of tube drawing is similar to wire drawing, but in this case tube is sized by shrinking a large diameter tube into a smaller one, by drawing the tube through a die.Schematically presents the set-ups of wire drawing and tube drawing.



(a) Wire Drawing and (b) Tube Drawing

DEFECTS IN DRAWING PROCESS

- Most of the defects in drawing process are similar to those observed in extrusion, specially centre cracking. The tendency of cracking increases with the increase in die angle.
- Another type of defect in drawing is the formation of seams, which are longitudinal scratches of folds in the material.

SHEET METAL PROCESSES

Sheet metal work processing is highly common in manufacturing sheet metal parts usingfrom sheet stock. The various sheet metal operations are performed on press machine ofrequired capacity using press tools or dies.

The dies may be single operation die or multioperationdies. A simple piercing, blanking and shearing die is shown,howeverthe basic sheet metal operations are described in the following lines.

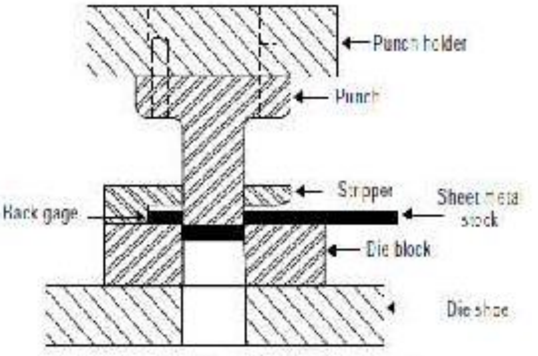
General Sheet Metal Operations

Shearing

It takes place when punch and die are used. The quality of the cut surface is greatlyinfluenced by the clearance between the two shearing edges. However, the basic shearingoperations are described in the following lines.

Cutting

It means severing a piece from a strip with a cut along a single line.



Typical simple press tool

Parting

It signifies that scrap is removed between the two pieces to part them.

Blanking

It means cutting a whole piece from sheet metal just enough scrap is left all around theopening to assure that the punch has metal to cut along its entire edge. The piece detachedfrom the strip is known as blank and is led for further operations. The remaining metal stripis scrap.

Punching

It is the operation of producing circular holes on a sheet metal by a punch and die. Thematerial punched out is removed as waste. Piercing, on the other hand, is the process ofproducing holes of any desired shape.

Notching

It is a process of removing metal to the desired shape from the side or edge of a sheetor strip.

Slitting

When shearing is conducted between rotary blades, the process is referred to as slitting.
It cuts the sheet metal lengthwise.

Nibbling

It is an operation of cutting any shape from sheet metal without special tools. It is done
On a nibbling machine.

Trimming

It is the operation of cutting away excess metal in a flange or flash from a piece.

Lancing

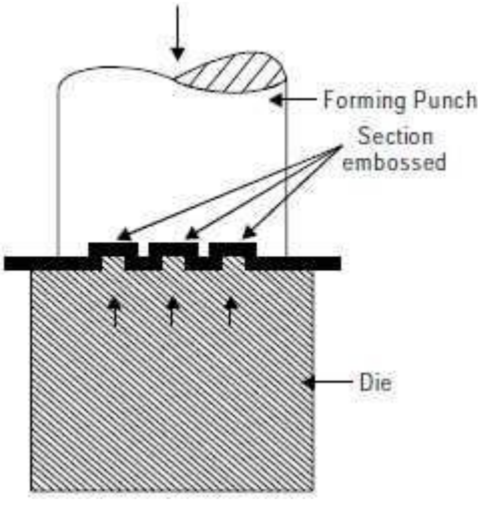
It makes a cut part way across a strip.

Forming

It is a metal working process in which the shape

COINING

Coining process used in cold workingoperations. It is basically a cold working operation,which is performed in dies where the metal blank isconfined and its lateral flow is restricted. It is mainlyused for production of important articles such as medals,coins, stickers and other similar articles, which possessshallow configurations on their surfaces.



The operationinvolves placing a metal slug in the die and applyingheavy pressure by the punch. The metal flows plasticallyand is squeezed to the shape between punch and thedie. The process, on account of the very high pressuresrequired, can be employed only for soft metals withhigh plasticity.